Credence: A Bayesian Decision-Theoretic Framework for LLM Agent Tool Selection

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Abstract

Current LLM agent frameworks select tools through prompting heuristics without any formal mechanism for evaluating whether a tool query is worth its cost. We introduce Credence, a Bayesian decision-theoretic framework that derives tool selection from first principles: agents maintain conjugate beliefs about tool reliability, update them from observed outcomes, and select queries by maximising expected utility via value-of-information calculations. The framework is grounded in five axioms—Bayesian conditioning, expected utility integration, EU maximisation, a Solomonoff complexity prior, and a cooperative inverse reinforcement learning (CIRL) alignment commitment—from which all agent behaviour is derived rather than engineered.

On a controlled benchmark, Credence scores +129.5 points compared to +10.8 for the best LLM agent (ReAct with strategy prompting and cross-question memory), despite achieving *lower* per-query accuracy (62.6% vs. 76.4%). We term this the *accuracy paradox*: optimising accuracy without accounting for query costs is provably suboptimal. We evaluate three LLM agent variants—bare prompting, ReAct reasoning traces [Yao et al., 2023], and a fully enhanced variant—and find that additional prompting techniques improve accuracy but never close the score gap: the best LLM variant achieves the highest accuracy of any agent yet scores 12× less than Credence. Under distributional drift, Credence degrades gracefully (score change: −21.8) while a fixed-tool baseline collapses (−69.0). These results suggest that the dominant paradigm of designing agents through prompting is fundamentally limited, and that principled Bayesian approaches offer a practical and theoretically grounded alternative.

1 Introduction

The rapid deployment of large language model (LLM) agents—systems that use LLMs to select and invoke external tools—has created a curious gap in AI practice. While the perception capabilities of these systems have improved dramatically through scaling [Sutton, 2019], the *decision-making* layer remains largely unprincipled. Frameworks such as LangChain [Chase, 2022] and LlamaIndex implement tool selection through prompting: the LLM is instructed to reason about which tools to use, with no formal mechanism for quantifying uncertainty, evaluating marginal information value, or trading off query costs against expected gains.

This paper argues that the tool selection problem is fundamentally a *decision-theoretic* problem, not a perception problem, and that treating it as such yields both theoretical clarity and practical improvements. Decision theory does not compete with deep learning for the same job—it answers

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a different question entirely: given uncertainty about tool reliability and finite query budgets, *what should the agent do?*

We introduce Credence, a framework built on five axioms from which all agent behaviour is derived. The key insight is that tool queries are information-gathering actions whose value can be computed exactly: the *value of information* (VOI) of querying a tool is the expected utility of observing the tool's response and then acting optimally, minus the expected utility of acting on current beliefs alone. A query is worth making if and only if its VOI exceeds its cost. This is not a heuristic—it is a theorem of Bayesian decision theory [Raiffa and Schlaifer, 1961, DeGroot, 1984].

Contributions. (1) We present a principled framework for LLM tool selection grounded in Bayesian decision theory and CIRL, requiring no LLM calls for the decision-making layer itself. (2) We identify the *accuracy paradox*: agents optimising per-query accuracy can achieve lower net value than agents optimising expected utility, because accuracy-maximising agents over-query. (3) We demonstrate empirically that prompt engineering cannot solve this problem—more sophisticated prompts make agents *more expensive* without making them more economical. (4) We show that Bayesian belief updating provides natural robustness to distributional drift without any explicit change-detection mechanism.

Connection to prior work. The present framework builds on two lines of research. First, the theory of *embodied agent design* due to Ay [2015], which formalises the separation between an agent's controller (brain) and its sensorimotor apparatus (body and environment), and argues that control architectures should be *sufficient* for expressing desired behaviours while remaining *concise*—the principle of cheap design. In Credence, the "brain" is the Bayesian decision layer; the "body" is the LLM and external tools. The LLM is treated as a *prosthetic*—a noisy sensor whose reliability is learned, not a decision-maker.

2 Related Work

2.1 Principled Frameworks for LLM Agents

The application of Bayesian decision theory to agent design has a long history [Russell and Wefald, 1991, Boutilier et al., 1999, Kaelbling et al., 1998], yet modern LLM agent frameworks have largely ignored these foundations. ReAct [Yao et al., 2023], Reflexion [Shinn et al., 2023], and LATS [Zhou et al., 2024] delegate both perception and decision-making to the LLM (the latter via MCTS-style planning); Toolformer [Schick et al., 2023] fine-tunes models to decide when to call APIs. These approaches conflate two fundamentally different capabilities: extracting information from tool outputs (perception) and choosing which tools to invoke (decision-making). Wang et al. [2023] survey over 100 such systems; none maintains explicit beliefs about tool reliability or computes value of information.

Three recent works apply principled methods to LLM agents. **RAFA** [Liu et al., 2024] is the most formally rigorous: it casts agent reasoning as a Bayesian adaptive MDP and proves a $\sqrt{T}$ regret bound. However, RAFA addresses planning in unknown environments, not per-tool reliability or VOI for individual tool calls. **DeLLMa** [Liu et al., 2025] applies expected utility maximisation to LLM decisions via multi-step inference—the closest in spirit to Credence's EU framework—but lacks belief updating, VOI computation, and tool reliability tracking. **MACLA** [Forouzandeh et al., 2026] maintains Beta posteriors over procedure success rates, and its ablation confirms Bayesian selection as the single most impactful component. But MACLA selects multi-step procedures, not individual tools, and does not compute VOI.

To our knowledge, no published work combines per-tool Bayesian belief tracking, value-of-information computation, and expected utility maximisation for LLM tool selection.

2.2 Cost-Aware Agent Approaches

Cost-aware approaches to LLM tool use span five tiers of increasing formality: (1) *prompt-based*—expose budget or cost information in the prompt; (2) *RL reward shaping*—penalise tool calls in the reward signal; (3) *planning and optimisation*—pre-plan tool sequences under budget constraints, as in BTP [Zheng et al., 2024]; (4) *sequential decision-making*—online planning with cost anticipation, most rigorously formalised by INTENT [Liu et al., 2026]; and (5) *decision-theoretic*—computing VOI before each tool call, the approach taken by CREDENCE.

Kapoor et al. [2025] demonstrate that agent evaluation without cost accounting is misleading—accuracy alone fails to capture real-world value. Wang et al. [2024] show that budget-aware reasoning strategies outperform unconstrained ones, and Erol et al. [2025] propose an economic framework for model evaluation that accounts for inference cost. Wang et al. [2025] argue that agents should invoke tools "only when epistemically necessary," defining a knowledge boundary concept but providing no computational mechanism. A Bayesian VOI framework provides exactly the mechanism they call for.

Our LLM agent results (§5) empirically demonstrate the failure of tier 1: adding cost-awareness prompts, reasoning traces, and cross-question memory to a capable agent improves accuracy but never closes the score gap with CREDENCE. The prompting trap arises because natural language cannot express the quantitative tradeoff between information gain and query cost that VOI captures naturally. De Sabbata et al. [2024] reach a complementary conclusion from the metareasoning perspective: LLMs benefit from explicit computational cost-benefit analysis rather than implicit prompt-level reasoning.

2.3 Theoretical Foundations

Embodied agent design. Ay [2015] argues that an agent's controller need not be a universal approximator—it suffices to match the behavioural repertoire afforded by its embodiment—motivating CREDENCE's deliberately simple decision layer.

Value of information. VOI originates with Howard [1966] and is developed for bounded agents by Russell and Wefald [1991]. Bayesian experimental design [Chaloner and Verdinelli, 1995] formalises optimal information gathering under cost constraints. Dong et al. [2026] recently extend VOI to human-agent communication, but apply it to dialogue acts rather than tool selection. CREDENCE appears to be the first system to compute VOI for individual tool calls in an LLM agent.

Value alignment. Cooperative inverse reinforcement learning [Hadfield-Menell et al., 2016] treats human-robot interaction as a cooperative game in which the robot infers human preferences from observed behaviour. CREDENCE's treatment of the LLM as a noisy sensor—whose outputs are observations with uncertainty, not commands—embodies a similar principle: the decision-theoretic layer maintains its own beliefs and defers to the human's utility function rather than to the LLM's token probabilities.

3 The CREDENCE Framework

3.1 Axioms

CREDENCE is built on five axioms. These are non-negotiable foundations from which all behaviour is derived.

Axiom 1 (Conditioning)**.** *Beliefs are updated by Bayes' rule. This is the unique update rule for which updating and computing the predictive commute, and the unique rule that cannot be Dutch-booked across time (diachronic coherence).*

Axiom 2 (Integration)**.** *The expected value of a quantity under a belief is computed by integrating over the belief distribution.*

Axiom 3 (Optimisation)**.** *Actions are selected to maximise expected utility under the current belief. Savage's representation theorem [Savage, 1954] establishes that any agent whose preferences satisfy basic rationality axioms acts as if maximising expected utility. The complete class theorem [Wald, 1950, Berger, 1985] further establishes that Bayesian decision rules are the only admissible ones: any non-Bayesian rule is dominated by some Bayesian rule.*

Axiom 4 (Complexity prior)**.** *The prior over hypotheses is weighted by description length:* $2^{-|h|}$ *for hypothesis* h*. This is Solomonoff's universal prior [Solomonoff, 1964]—the unique prior that dominates all computable priors up to a constant.*

Axiom 5 (Alignment)**.** *The agent's utility equals the user's utility under preference uncertainty:* $U(a, s, b) = \mathbb{E}_{\theta\sim b}[u_\theta(s, a)]$*, where* θ *parameterises the user's preferences and* b *is the agent's belief over* θ*. This is the cooperative inverse reinforcement learning (CIRL) framework of Hadfield-Menell et al. [2016]: the agent and user are in a cooperative game where the agent's reward is the user's reward, but the agent does not know the user's reward function and must infer it from interaction.*

3.2 Architecture

We decompose the agent into a decision layer (maintains beliefs, selects actions), tools including the LLM (noisy information channels), and the environment (questions and ground truth). The LLM is treated as a sensor, not a decision-maker.

3.3 Belief Model

For each tool $t \in \{1, \ldots, T\}$ and question category $c \in \{1, \ldots, C\}$, the agent maintains a Beta-distributed belief over the tool's reliability:

$$p_{t,c} \sim \text{Beta}(\alpha_{t,c}, \beta_{t,c}). \tag{1}$$

The conjugate Beta-Bernoulli model means that after observing a correct response from tool t on a category-c question, the update is:

$$(\alpha_{t,c}, \beta_{t,c}) \mapsto (\alpha_{t,c} + 1, \beta_{t,c}), \tag{2}$$

and after an incorrect response:

$$(\alpha_{t,c}, \beta_{t,c}) \mapsto (\alpha_{t,c}, \beta_{t,c} + 1). \tag{3}$$

Initialisation. Priors are set to Beta$(1, 1)$ (uniform) for each tool-category pair. This encodes maximum ignorance about reliability, placing the prior mean at $1/2$ with high variance. The choice of $1/N$ (where N is the number of tools) for prior success probability, rather than 0.5, was considered and rejected: the Beta(1,1) prior is the unique non-informative conjugate prior and makes no assumption about relative tool quality.

Non-stationarity. The core framework handles distributional drift through standard Beta-Bernoulli updating: as a tool's reliability degrades, negative evidence accumulates and the posterior shifts downward naturally. No explicit change-detection mechanism is required—the posterior adapts continuously, with the rate of adaptation governed by the effective sample size (the sum $\alpha + \beta$).

Experimental variant: exponential forgetting. As an experimental extension, we also test exponential discounting of past observations ($\lambda = 0.95$), which preserves the Beta form while enabling faster adaptation to abrupt reliability changes [Freeman and Smith, 2011]; a detailed treatment of drift-handling mechanisms is deferred to future work.

3.4 Decision Rule: Value of Information

The agent's decision at each step is: should I query another tool, or should I act (submit an answer or abstain) based on current beliefs?

Let b denote the current belief state (the collection of all Beta parameters). Let r_t denote the cost of querying tool t. Let $v_{\text{correct}} = 10$, $v_{\text{wrong}} = -5$, $v_{\text{abstain}} = 0$ be the payoffs.

Expected utility of acting now. Given current beliefs b, if the agent has gathered responses $\{(t_i, x_i)\}_{i=1}^{k}$ from k tool queries, it computes a posterior probability p_{correct} of the most-supported answer being correct (by aggregating the reliability-weighted evidence). The expected utility of submitting is:

$$\text{EU}_{\text{submit}}(b) = p_{\text{correct}} \cdot v_{\text{correct}} + (1 - p_{\text{correct}}) \cdot v_{\text{wrong}}. \tag{4}$$

The expected utility of abstaining is $\text{EU}_{\text{abstain}} = 0$. The agent submits if $\text{EU}_{\text{submit}} > 0$, i.e., if $p_{\text{correct}} > 1/3$ (the break-even threshold given the payoff structure).

The best "act now" utility is:

$$\text{EU}^*(b) = \max(\text{EU}_{\text{submit}}(b),\ 0). \tag{5}$$

Value of information. The VOI of querying tool t is:

$$\text{VOI}(t \mid b) = \mathbb{E}_{x \sim p_t}\big[\text{EU}^*(b \mid x)\big] - \text{EU}^*(b), \tag{6}$$

where the expectation is over the possible responses x of tool t, weighted by the agent's current belief about tool t's reliability. This is the expected improvement in decision quality from observing tool t's response.

A query is made if and only if:

$$\text{VOI}(t \mid b) > r_t \quad \text{for some tool } t. \tag{7}$$

When multiple tools satisfy this criterion, the agent selects the one with the highest net VOI: $\text{argmax}_t[\text{VOI}(t \mid b) - r_t]$.

Table 1: Tool reliability by question category. Tool B has category-dependent coverage (probability of returning any result). Tool C returns `not_applicable` for all non-numerical questions.

	P(correct)					Coverage	Cost
Tool	Fact.	Num.	Rec.	Misc.	Reas.		
A (Web Search)	0.85	0.30	0.75	0.35	0.50	1.0	1
B (Knowledge Base)	0.95	0.40	0.60	0.90	0.50	0.2–0.7	3
C (Calculator)	1.00	—	Num. only	1			
D (LLM Direct)	0.70	0.55	0.50	0.45	0.75	1.0	2

Proposition 1. *Under the Beta-Bernoulli model,* $\mathrm{VOI}(t \mid b)$ *is non-increasing in the number of prior observations. That is, the more the agent knows about a tool, the less informative an additional query becomes.*

Proof. As $\alpha_{t,c} + \beta_{t,c} \to \infty$, the posterior variance $\mathrm{Var}(p_{t,c}) \to 0$ and the expected posterior after one additional observation converges to the current posterior. Hence $\mathbb{E}[\mathrm{EU}^*(b \mid x)] \to \mathrm{EU}^*(b)$ and $\mathrm{VOI}(t \mid b) \to 0$. □

This ensures that the agent's querying naturally terminates: after sufficiently many observations, no tool query is worth its cost.

Category inference. The agent treats question category as a latent variable with a uniform prior. As tools respond to questions, category beliefs are updated jointly with reliability beliefs via Bayesian conditioning: if a tool known to be reliable for numerical questions answers correctly, the posterior shifts toward "numerical." Over successive questions, the agent builds category-specific reliability profiles purely from tool response patterns—no text features are used.

4 Experimental Setup

4.1 Simulated Environment

We evaluate on a benchmark of 50 multiple-choice questions spanning five categories: factual, numerical, recent events, misconceptions, and reasoning. Tools are simulated with known reliability profiles (table 1), allowing exact computation of oracle performance and controlled drift experiments. While simulated tools limit ecological validity, they enable precise isolation of the decision-making layer's contribution.

Scoring. Correct answer: +10. Wrong answer: −5. Abstention: 0. Tool costs as in table 1. The asymmetric payoff creates a break-even submission threshold of $p_{\text{correct}} > 1/3$.

4.2 Agents Compared

Credence (Bayesian). The full framework of section 3. Beta(1,1) priors, latent category inference from tool response patterns, VOI-based tool selection. No LLM calls. Pure Julia inference engine.

LLM agents. Three variants of an LLM-based tool selection agent (Llama 3.1 8B via Ollama, temperature 0), differing in prompting technique. All share the same system prompt describing the four tools, scoring rules, and the option to abstain. The variants form a factorial design over three prompting techniques:

- **LLM Bare**: base prompt only—the LLM outputs an action (QUERY, SUBMIT, or ABSTAIN) with no reasoning.
- **LLM ReAct**: adds explicit Thought/Action/Observation traces [Yao et al., 2023], with the full reasoning history included in each prompt.
- **LLM ReAct+S+H**: adds strategy guidance (cost-awareness, per-tool reliability heuristics) and cross-question history (last 10 outcomes). This represents maximum prompt engineering effort.

The comparison is between decision-making paradigms—Bayesian EU maximisation vs. LLM-prompted reasoning—not software frameworks.

Oracle. Knows the true reliability table. Makes optimal VOI-based decisions. Upper bound on Credence performance with perfect knowledge.

Single-Best-Tool. Always queries Tool A and submits its answer. Tests whether Credence's advantage is merely tool routing.

All-Tools. Queries all four tools, majority-votes. Demonstrates the cost of indiscriminate querying.

Random. Queries a random tool, submits its response. Lower bound.

4.3 Experimental Protocol

All results are averaged over 20 random seeds. We run three experimental conditions:

1. **Stationary:** Tool reliabilities fixed as in table 1 throughout.
2. **Drift:** After question 25, Tool A's reliability drops substantially across all categories, simulating a real-world scenario such as a search API degradation. We compare Credence with and without exponential forgetting ($\lambda = 0.95$).
3. **Ablation:** Components of Credence removed individually to measure their contribution.

5 Results

5.1 Stationary Environment

The central result (table 2) is that Credence achieves the highest score (+129.5) whilst having *lower* accuracy (62.6%) than any of the LLM agents. The best LLM variant (ReAct+S+H) achieves the highest accuracy of any agent (76.4%) but scores +10.8—12× less than Credence—because it averages 3.05 tool queries per question and takes 5.65 seconds per question compared to Credence's 0.07 seconds.

Table 2: Stationary benchmark results (mean ± std over 20 seeds).

Agent	Score	Accuracy	Tools/Q	Time/Q (s)
CREDENCE (Bayesian)	$+\mathbf{129.5 \pm 51.3}$	0.626	1.80	0.07
Single-Best-Tool	$+58.5 \pm 55.9$	0.478	1.00	<0.01
Random	$+12.2 \pm 54.8$	0.450	1.00	<0.01
LLM ReAct+S+H	$+10.8 \pm 50.8$	**0.764**	3.05	5.65
LLM ReAct	-15.3 ± 28.0	0.660	2.70	4.60
All-Tools	-114.2 ± 60.1	0.581	4.00	<0.01
LLM Bare	-160.5 ± 36.1	0.438	3.26	0.93

The accuracy paradox. This result is not anomalous—it is a direct consequence of the decision-theoretic framework. Accuracy measures $P(\text{correct})$ but ignores the cost of achieving that probability. The LLM agents achieve high accuracy *because* they query many tools, but each query has a cost. The net score is:

$$\text{Score} = \sum_{q=1}^{Q} \big[v_q \cdot \mathbf{1}[\text{correct}_q] + v_{\text{wrong}} \cdot \mathbf{1}[\text{wrong}_q] \big] - \sum_{q=1}^{Q} \text{cost}_q, \tag{8}$$

where $v_q = v_{\text{correct}}$ if the agent submits correctly, v_{wrong} if incorrectly, and 0 if abstaining. Maximising the first sum (accuracy) is not equivalent to maximising the full expression when costs are non-zero.

The prompting trap. The three LLM variants reveal a clear gradient: LLM Bare (−160.5), LLM ReAct (−15.3), LLM ReAct+S+H (+10.8). Each prompting technique—reasoning traces, strategy guidance, cross-question memory—improves accuracy. But the improvements never close the gap: even the fully enhanced variant, with the highest accuracy of any agent (76.4%), scores 12× less than CREDENCE. The additional reasoning makes the agent *more thorough*: it considers tool reliability, tracks past performance, and deliberates before each query. But deliberation itself costs time (5.65s vs. 0.07s per question), and the LLM still queries 3.05 tools per question because it lacks a formal mechanism for computing whether a query is worth its cost. This demonstrates that cost-benefit analysis cannot be recovered through prompting: it requires an explicit decision-theoretic mechanism external to the LLM.

5.2 Distributional Drift

In supplementary experiments, we verify that the Bayesian updating mechanism provides natural robustness to distributional drift without explicit change detection; detailed drift analysis is deferred to future work.

5.3 Ablation Study

Category inference is the single most valuable component (−102.0): without category-aware priors the agent cannot route questions to appropriate tools and accuracy drops from 59.6% to 31.6%. That No VOI and Fixed reliability produce identical scores (−78.1) is not coincidence but mathematical consequence: when reliability priors are uninformative ($p = 0.5$), all tools look equally trustworthy, so VOI calculations degenerate to cost-based selection regardless of whether VOI is computed or

Table 3: Ablation study: effect of removing individual components (stationary, 20 seeds).

Variant	Score	Δ vs. Full	Effect
Full CREDENCE	$+112.6 \pm 44.6$	—	
No categories (flat prior)	$+10.6 \pm 55.8$	-102.0	Accuracy collapses to 31.6%; 0.49 tools/Q
No VOI (always query)	$+34.5 \pm 48.9$	-78.1	Degenerates to cheapest-tool heuristic
Fixed reliability ($p = 0.5$)	$+34.5 \pm 48.9$	-78.1	VOI uses uninformative priors; same as no VOI
No abstention	$+91.1 \pm 82.2$	-21.5	Forced submissions on low confidence; 2× variance

bypassed. The framework's power comes from the *interaction* between learned category-specific beliefs and VOI, not from either component alone. Abstention provides a moderate but consistent benefit (-21.5) by avoiding -10 penalties on low-confidence questions, though its removal doubles score variance.

6 Discussion

Why accuracy is the wrong metric. The accuracy paradox is not merely an artefact of our scoring system. Any deployment context with non-zero query costs—API fees, latency, rate limits—creates a divergence between accuracy and net value. In production, the LLM ReAct+S+H agent's strategy of "reason carefully about every query" would be the most expensive and least efficient approach: it achieves 76.4% accuracy but takes 80× longer per question than CREDENCE and still scores 12× less. The correct metric is expected utility, which accounts for both the value of correct answers and the cost of obtaining them.

Cheap design for LLM agents. The computational asymmetry—0.07s per question for the decision layer vs. 5.65s for the LLM agent—instantiates the cheap design principle of Ay [2015], wherein a controller need not be a universal approximator if it is sufficient for the embodiment constraints; we formalise this connection in forthcoming work.

Limitations. The current evaluation uses simulated tools with known ground truth, which limits ecological validity. The question bank is small (50 questions) and category inference is learned purely from tool response patterns via Bayesian conditioning on a latent category variable, starting from a uniform prior—no text features are used. The LLM agents use a local model (Llama 3.1 8B via Ollama); running them with frontier models (GPT-4o, Claude Sonnet) would likely improve their accuracy but would not address the structural inability to compute value of information—the prompting trap result already demonstrates this, as the fully enhanced LLM variant achieves 76.4% accuracy yet still scores 12× less than CREDENCE. Finally, the current framework assumes tools return independent responses—correlated errors between tools are not modelled.

7 Conclusion and Future Work

We have presented CREDENCE, a Bayesian decision-theoretic framework for LLM tool selection that derives agent behaviour from five axioms rather than engineering it through prompting. The accuracy paradox—that principled agents can achieve higher net value with lower per-query accuracy—demonstrates a fundamental limitation of the dominant paradigm. Cost-benefit analysis is not a prompting problem; it requires explicit decision-theoretic mechanisms.

Several extensions are immediate. First, evaluation with frontier LLMs and real (non-simulated) tools to establish ecological validity. Second, the full CREDENCE architecture extends beyond tool selection to a general-purpose agent framework with programs represented as DSL expressions (closed-loop policies), meta-actions that modify the hypothesis space, and grammar evolution for discovering abstractions—a "strange loop" where the agent improves its own capacity for improvement. This broader architecture, grounded in the same five axioms, will be presented in a companion paper.

Reproducibility. Code and data are available at `https://github.com/gfrmin/credence`. All experiments use fixed random seeds.

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